

CYBER SECURITY ANALYST INTERNSHIP PROGRAM



WEEK 3 ASSIGNMENT

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Task 1: Protocols and Port Numbers

Objective

The purpose of this task is to learn about commonly used network protocols, their default port numbers, the transport protocol they use (TCP or UDP), and whether they provide secure or insecure communication.

Protocols and Port Numbers

1. HTTP (Hypertext Transfer Protocol)

- Port Number: 80
- Protocol: TCP
- Purpose: HTTP allows users to open and view web pages on the Internet. It transfers website content between the browser and the web server without encrypting the data.
- Security: Insecure

2. HTTPS (Hypertext Transfer Protocol Secure)

- Port Number: 443
- Protocol: TCP
- Purpose: HTTPS provides a secure connection between a user and a website by encrypting all transmitted information, making online communication more secure.
- Security: Secure

3. FTP (File Transfer Protocol)

- Port Number: 21
- Protocol: TCP
- Purpose: FTP is mainly used to upload, download, and manage files on a remote server over a network.
- Security: Insecure

4. SFTP (Secure File Transfer Protocol)

- Port Number: 22
- Protocol: TCP
- Purpose: SFTP transfers files securely through an encrypted SSH connection, protecting data from unauthorized access.
- Security: Secure

5. SSH (Secure Shell)

- Port Number: 22
- Protocol: TCP

- Purpose: SSH allows secure remote access to computers and servers, enabling administrators to manage systems safely.
- Security: Secure

6. Telnet

- Port Number: 23
- Protocol: TCP
- Purpose: Telnet provides remote access to another computer through a command-line interface but sends data in plain text.
- Security: Insecure

7. SMTP (Simple Mail Transfer Protocol)

- Port Number: 25
- Protocol: TCP
- Purpose: SMTP is responsible for sending outgoing emails from users and transferring emails between mail servers.
- Security: Insecure

8. DNS (Domain Name System)

- Port Number: 53
- Protocol: TCP/UDP
- Purpose: DNS helps users access websites by converting easy-to-read domain names into numerical IP addresses.
- Security: Insecure

9. DHCP (Dynamic Host Configuration Protocol)

- Port Number: 67/68
- Protocol: UDP
- Purpose: DHCP automatically provides network settings such as IP address, subnet mask, gateway, and DNS server to devices connected to a network.
- Security: Insecure

10. TFTP (Trivial File Transfer Protocol)

- Port Number: 69
- Protocol: UDP
- Purpose: TFTP is commonly used to transfer small files like firmware updates and network device configuration files.
- Security: Insecure

11. POP3 (Post Office Protocol Version 3)

- Port Number: 110
- Protocol: TCP

- Purpose: POP3 retrieves emails from a mail server and stores them on the user's local device for offline access.
- Security: Insecure

12. IMAP (Internet Message Access Protocol)

- Port Number: 143
- Protocol: TCP
- Purpose: IMAP lets users read and organize emails while keeping them stored on the mail server, allowing access from multiple devices.
- Security: Insecure

13. SNMP (Simple Network Management Protocol)

- Port Number: 161
- Protocol: UDP
- Purpose: SNMP is used to monitor the health, performance, and status of network devices from a central management system.
- Security: Insecure

14. LDAP (Lightweight Directory Access Protocol)

- Port Number: 389
- Protocol: TCP/UDP
- Purpose: LDAP is used to access and manage directory services, including user accounts and authentication information in organizations.
- Security: Insecure

15. RDP (Remote Desktop Protocol)

- Port Number: 3389
- Protocol: TCP
- Purpose: RDP allows a user to remotely connect to and operate a Windows computer through a graphical desktop interface.
- Security: Secure

Task 2: HTTP vs HTTPS Analysis

Objective

The aim of this task is to compare HTTP and HTTPS by analyzing different websites. This helps in understanding website security, encrypted communication, and automatic redirection from HTTP to HTTPS.

1. Google (google.com)

- Protocol Used: HTTPS
- HTTP Redirect to HTTPS: Yes
- Connection Secure: Yes

Explanation: Google uses HTTPS to protect user searches and account information. If a user enters the HTTP version, it automatically switches to HTTPS for a secure connection.

2. Instagram (instagram.com)

- Protocol Used: HTTPS
- HTTP Redirect to HTTPS: Yes
- Connection Secure: Yes

Explanation: Instagram encrypts all communication between the user and the server. This helps protect login details, messages, and uploaded content.

3. X (x.com)

- Protocol Used: HTTPS
- HTTP Redirect to HTTPS: Yes
- Connection Secure: Yes

Explanation: X uses HTTPS to keep user accounts and posts secure. The website automatically redirects users to the encrypted version.

4. Reddit (reddit.com)

- Protocol Used: HTTPS
- HTTP Redirect to HTTPS: Yes
- Connection Secure: Yes

Explanation: Reddit uses HTTPS to provide secure browsing and protect user accounts and community interactions.

5. Netflix (netflix.com)

- Protocol Used: HTTPS
- HTTP Redirect to HTTPS: Yes
- Connection Secure: Yes

Explanation: Netflix uses HTTPS to secure user accounts, subscriptions, and streaming sessions while protecting personal information.

6. Canva (canva.com)

- Protocol Used: HTTPS
- HTTP Redirect to HTTPS: Yes
- Connection Secure: Yes

Explanation: Canva protects user designs and login information using HTTPS encryption and redirects all HTTP requests to HTTPS.

7. Zoom (zoom.us)

- Protocol Used: HTTPS
- HTTP Redirect to HTTPS: Yes
- Connection Secure: Yes

Explanation: Zoom uses HTTPS to secure access to meetings, user accounts, and communication with its web services.

8. Dropbox (dropbox.com)

- Protocol Used: HTTPS
- HTTP Redirect to HTTPS: Yes
- Connection Secure: Yes

Explanation: Dropbox uses HTTPS to protect uploaded files and user accounts from unauthorized access during transmission.

9. PayPal (paypal.com)

- Protocol Used: HTTPS
- HTTP Redirect to HTTPS: Yes
- Connection Secure: Yes

Explanation: PayPal uses HTTPS to secure financial transactions and protect sensitive customer information such as payment details.

10. Yahoo (yahoo.com)

- Protocol Used: HTTPS
- HTTP Redirect to HTTPS: Yes
- Connection Secure: Yes

Explanation: Yahoo uses HTTPS to secure email services, user accounts, and other online services by encrypting network traffic.

Task 3: Kali Linux Network Reconnaissance

The objective of this task is to perform basic network reconnaissance in Kali Linux and identify active hosts, open ports, and running services within a network. Reconnaissance is the first step of a security assessment and helps in understanding the network environment.

Command 1: Check Connectivity

Ping Google.com

ScreenShot :

```

(kali@kali)-[~]
$ ping google.com
PING google.com (142.250.184.110) 56(84) bytes of data.
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=1 ttl=255 time
=33.9 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=2 ttl=255 time
=36.6 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=3 ttl=255 time
=40.6 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=4 ttl=255 time
=33.9 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=5 ttl=255 time
=34.9 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=6 ttl=255 time
=39.3 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=7 ttl=255 time
=34.0 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=8 ttl=255 time
=34.6 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=9 ttl=255 time
=37.8 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=10 ttl=255 tim
e=35.6 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=11 ttl=255 time=34.0 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=12 ttl=255 time=37.2 ms
64 bytes from mil41s04-in-f14.1e100.net (142.250.184.110): icmp_seq=13 ttl=255 time=35.2 ms
^C
--- google.com ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12014ms
rtt min/avg/max/mdev = 33.867/35.964/40.554/2.102 ms

```

Result Analysis

- The domain google.com was successfully resolved to the IP address 142.250.184.110.
- A total of 13 ICMP packets were transmitted.
- All 13 packets were received successfully.
- Packet Loss: 0%
- Minimum RTT: 33.867 ms
- Average RTT: 35.964 ms
- Maximum RTT: 40.554 ms
- Standard Deviation: 2.102 ms

Command 2: Scan latency.nmap.org

latency.nmap.org

```

(kali@kali)-[~]
$ nmap latency.nmap.org
Starting Nmap ( https://nmap.org ) at 2026-07-17 19:22 UTC
Nmap scan report for latency.nmap.org
Host is up (0.15s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
31337/tcp  open  Elite

Nmap done: 1 IP address (1 host up) scanned in 97.59 seconds

```

Here are the details and a short explanation of the Nmap scan shown in the image:

- Target Host: **latency.nmap.org**
- Host Status: Up (responding to scans)
- Network Latency: 0.15 seconds
- Open Ports Discovered:
 - 22/tcp: Open, running SSH (Secure Shell for remote administration)
 - 80/tcp: Open, running HTTP (Web server protocol)
 - 31337/tcp: Open, running Elite (often used as a default testing port or associated with older backdoor Trojans)
- Scan Summary: 1 IP address (1 host up) scanned successfully.
- Total Duration: 97.59 seconds

Command 2: Scan Local Ip

```
(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 83962sec preferred_lft 83962sec
    inet6 fd17:625c:f037:2:b1e7:aa12:fc67:97fd/64 scope global dynamic noprefixroute
        valid_lft 86392sec preferred_lft 14392sec
    inet6 fe80::f63c:3c82:8a8f:1aa5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali㉿kali)-[~]
$ nmap -sV 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-16 19:38 -0400
Nmap scan report for 10.0.2.15
Host is up.
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 202.24 seconds

(kali㉿kali)-[~]
$
```

Result Summary

- Local IP Address: 10.0.2.15
- Scan Command: nmap -sV 10.0.2.15
- Host Status: Host is up

Response Time (ms)

- The screenshot does not display a specific response time in milliseconds.
- The host responded successfully, which confirms that the system is reachable.

TTL Value

- The TTL value is not shown in the Nmap scan output.
- TTL values are normally visible in ping results, not in this Nmap output.

Open Ports:

- None detected
- All 1000 scanned TCP ports were reported as filtered.
- This means the ports did not respond to the scan request

Services Running on Open Ports

- No open ports were identified.
- Therefore, no running services were detected on the scanned ports.

Potential Security Risk

Since no open ports were detected, there is no immediate exposed service visible in this scan. Filtered ports may indicate that a firewall is blocking scan requests, which generally improves security by limiting external access.

Task 4: What Happens When You Type a URL and Press Enter?

The objective of this task is to explain the sequence of networking and security operations that occur when a user enters a website address in a browser. Understanding this process is important for cyber security analysts because each stage can be monitored, secured, or targeted by attackers.

1. Domain Name Resolution

When a URL such as `www.google.com` is entered, the browser must determine the server's IP address. It queries a DNS resolver, which returns the appropriate IP address associated with the domain.

Example:

`www.google.com` → `142.250.202.46`

Cyber Security Note: DNS traffic is often monitored because attackers may use malicious domains or DNS-based communication channels.

2. Network Route Identification

After receiving the IP address, the operating system determines how to reach the destination through the local gateway and internet routing infrastructure. This allows packets to be forwarded toward the target web server.

Cyber Security Note: Incorrect routing or spoofed gateway information can redirect traffic to unauthorized systems.

3. Connection Establishment

A reliable TCP connection is created using the three-way handshake process.

- SYN: Client requests a connection.
- SYN-ACK: Server acknowledges the request.
- ACK: Client confirms the connection.

Cyber Security Note: Security tools often analyze handshake behavior to detect scanning activity or denial-of-service attempts.

4. Secure Session Negotiation

For HTTPS websites, the browser and server negotiate encryption parameters through a TLS handshake.

During this stage:

- The server provides its digital certificate.
- The browser validates the certificate authority.
- Encryption algorithms are selected.
- Session keys are generated for secure communication.

Cyber Security Note: Certificate validation helps prevent man-in-the-middle attacks and unauthorized interception.

5. Web Request Transmission

Once the secure session is established, the browser sends an HTTP/HTTPS request to retrieve the requested resource.

Example:

GET / HTTP/1.1

Host: www.google.com

Cyber Security Note: Web application firewalls (WAFs) inspect requests for malicious payloads such as SQL injection or cross-site scripting attempts.

6. Server Processing and Response

The server processes the request and returns a response containing the requested content and an HTTP status code.

Common status codes include:

- 200 OK – Request processed successfully.
- 301 Moved Permanently – Resource redirected.
- 403 Forbidden – Access denied.
- 404 Not Found – Resource unavailable.
- 500 Internal Server Error – Server-side issue.

Cyber Security Note: Repeated error responses can indicate probing or unauthorized access attempts.

7. Browser Rendering and Resource Loading

The browser interprets the received content and constructs the webpage.

It performs the following actions:

- Parses the HTML document.
- Applies CSS styling.
- Executes JavaScript code.
- Downloads images, fonts, and additional resources.
- Displays the final webpage to the user.

Task 5: Networking Cheat Sheet

Protocol Name	Port Number	TCP/UDP	Purpose	Security Risk	Secure Alternative
HTTP	80	TCP	Web browsing	Data transmitted in plain text	HTTPS
HTTPS	443	TCP	Secure web browsing	Low	—
FTP	21	TCP	File transfer	Credentials sent unencrypted	SFTP / FTPS
SSH	22	TCP	Secure remote access	Low	—
Telnet	23	TCP	Remote terminal access	Plain-text communication	SSH

Protocol Name	Port Number	TCP/UDP	Purpose	Security Risk	Secure Alternative
SMTP	25	TCP	Email sending	Email content may be exposed	SMTPS
DNS	53	TCP/UDP	Domain name resolution	DNS spoofing / interception	DNS over HTTPS (DoH)
DHCP	67/68	UDP	IP address assignment	Rogue DHCP attacks	DHCP Snooping
POP3	110	TCP	Email retrieval	Unencrypted email access	POP3S
IMAP	143	TCP	Email synchronization	Unencrypted email access	IMAPS
SNMP	161	UDP	Network management	Weak community strings	SNMPv3
LDAP	389	TCP/UDP	Directory services	Credentials may be exposed	LDAPS
RDP	3389	TCP	Remote desktop access	Brute-force attacks	RDP with MFA/VPN

Task 6: Kali Linux Web Enumeration

Website : **Google** (google.com)

whatweb google.com

whois google.com

```
(kali@kali)-[~]
$ whatweb google.com
http://google.com [301 Moved Permanently] Country[UNITED STATES][us], HTTPServer[gws], IP[142.250.184.14], RedirectLocation[http://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
https://google.com [301 Moved Permanently] Country[UNITED STATES][us], HTTPServer[gws], IP[142.250.184.14], RedirectLocation[https://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
http://www.google.com/ [200 OK] Cookies[AEC,NID,Secure-STRP], Country[UNITED STATES][us], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.155.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
https://www.google.com/ [200 OK] Cookies[AEC,NID,Secure-STRP], Country[UNITED STATES][us], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.155.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only,accept-ch,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]

(kali@kali)-[~]
$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-17T00:03:30Z <<<
```

Results :

- Target Website: google.com
- HTTP Status: 301 Moved Permanently (redirects to www.google.com)
- HTTPS Status: 200 OK
- Country: United States (US)
- Web Server: gws (Google Web Server)
- Detected Technology: HTML5
- IP Addresses Detected: 142.250.184.14 and 142.251.155.119

Security-Related Headers Detected

- X-Frame-Options: SAMEORIGIN → Helps prevent clickjacking attacks.
- X-XSS-Protection: 0 → Browser-based XSS protection is disabled.
- Content-Security-Policy-Report-Only → Used for monitoring content security policy violations.
- HttpOnly Cookies: AEC and NID cookies are marked as HttpOnly, reducing the risk of JavaScript-based cookie theft.

WhatWeb Key Observations

- HTTP traffic is automatically redirected to HTTPS.
- The website uses encrypted HTTPS communication.
- Google has implemented multiple security headers.
- Cookies are configured with secure attributes such as Secure and HttpOnly.

Whois Lookup Details

- Domain Name: GOOGLE.COM
- Registrar: MarkMonitor Inc.
- Registrar WHOIS Server: whois.markmonitor.com
- Creation Date: 1997-09-15
- Updated Date: 2019-09-09
- Expiry Date: 2028-09-14
- DNSSEC: Unsigned

Name Servers

- NS1.GOOGLE.COM

- NS2.GOOGLE.COM
- NS3.GOOGLE.COM
- NS4.GOOGLE.COM

Domain Protection Status

The Whois output shows several protection flags:

- clientDeleteProhibited
- clientTransferProhibited
- clientUpdateProhibited
- serverDeleteProhibited
- serverTransferProhibited
- serverUpdateProhibited

These restrictions help prevent unauthorized deletion, transfer, or modification of the domain.

Website : GitHub (github.com)

whatweb github.com

whois github.com

```
(kali@kali)~$ whatweb github.com
https://github.com [301 Moved Permanently] Country[UNITED STATES][US], IP[20.207.73.82], RedirectLocation[https://github.com/]
https://github.com [200 OK] Content-Language[en-US], Cookies[gh_sess,octo_logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object][1403480693436526], OpenSearch[opensearch.xml], Script[application/json,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]
https://github.com/ [200 OK] Content-Language[en-US], Cookies[gh_sess,octo_logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object][1403480693436526], OpenSearch[opensearch.xml], Script[application/json,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referrer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]

(kali@kali)~$ whois github.com
Domain Name: GITHUB.COM
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-07T09:16:32Z
Creation Date: 2007-10-09T18:20:50Z
Registry Expiry Date: 2026-10-09T18:20:50Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: DNS1.P08.NSONE.NET
Name Server: DNS2.P08.NSONE.NET
Name Server: DNS3.P08.NSONE.NET
Name Server: DNS4.P08.NSONE.NET
Name Server: NS-1283.AWSDNS-32.ORG
Name Server: NS-1707.AWSDNS-21.CO.UK
Name Server: NS-421.AWSDNS-52.COM
Name Server: NS-520.AWSDNS-01.NET
DNSSEC: unsigned
```

Results:

- Target Website: github.com
- HTTP Status: 301 Moved Permanently (redirects to HTTPS)
- HTTPS Status: 200 OK
- Country: United States (US)

- Web Server: github.com
- IP Address: 20.207.73.82
- Detected Technology: HTML5
- Open Graph Protocol: Detected
- OpenSearch: /opensearch.xml
- Scripts Detected: application/json, module

Security-Related Headers Detected

- Strict-Transport-Security (HSTS) → Forces browsers to use HTTPS.
- X-Frame-Options: deny → Prevents the site from being loaded inside an iframe.
- X-XSS-Protection: 0 → Browser-based XSS filter disabled.
- Content-Security-Policy → Helps restrict unauthorized scripts and resources.
- X-Content-Type-Options → Prevents MIME-type sniffing.
- Referrer-Policy → Controls how referral information is shared.
- HttpOnly Cookies: _gh_sess, octo, and logged_in cookies are protected from JavaScript access.

WhatWeb Key Observations

- HTTP requests are automatically redirected to HTTPS.
- GitHub enforces secure communication using HSTS.
- Multiple security headers are implemented to reduce web-based attacks.
- Session cookies use the HttpOnly attribute for additional protection.

Whois Lookup Details

- Domain Name: GITHUB.COM
- Registrar: MarkMonitor Inc.
- Registrar WHOIS Server: whois.markmonitor.com
- Creation Date: 2007-10-09
- Updated Date: 2024-09-07
- Expiry Date: 2026-10-09

- DNSSEC: Unsigned

Name Servers

- DNS1.P08.NSONE.NET
- DNS2.P08.NSONE.NET
- DNS3.P08.NSONE.NET
- DNS4.P08.NSONE.NET
- NS-1283.AWSDNS-32.ORG
- NS-1707.AWSDNS-21.CO.UK
- NS-421.AWSDNS-52.COM
- NS-520.AWSDNS-01.NET

Domain Protection Status

The Whois output shows the following protection flags:

- clientDeleteProhibited
- clientTransferProhibited
- clientUpdateProhibited

These restrictions help protect the domain from unauthorized deletion, transfer, or modification.

Google (google.com) – Interesting Points

- Automatic HTTPS Redirection: Google automatically redirects all HTTP requests to HTTPS, ensuring encrypted communication for users.
- Strong Domain Protection: The Whois record shows multiple protection statuses such as clientTransferProhibited and serverDeleteProhibited, which help prevent unauthorized changes to the domain.

GitHub (github.com) – Interesting Points

- HSTS Enabled: GitHub uses Strict-Transport-Security (HSTS), forcing browsers to always use HTTPS and reducing the risk of downgrade attacks.
- Multi-Provider DNS Infrastructure: GitHub uses both NS1 and AWS Route 53 name servers, which improves reliability and availability of its domain services.